

TrashUQ

Federated Learning for Intelligent Waste Monitoring

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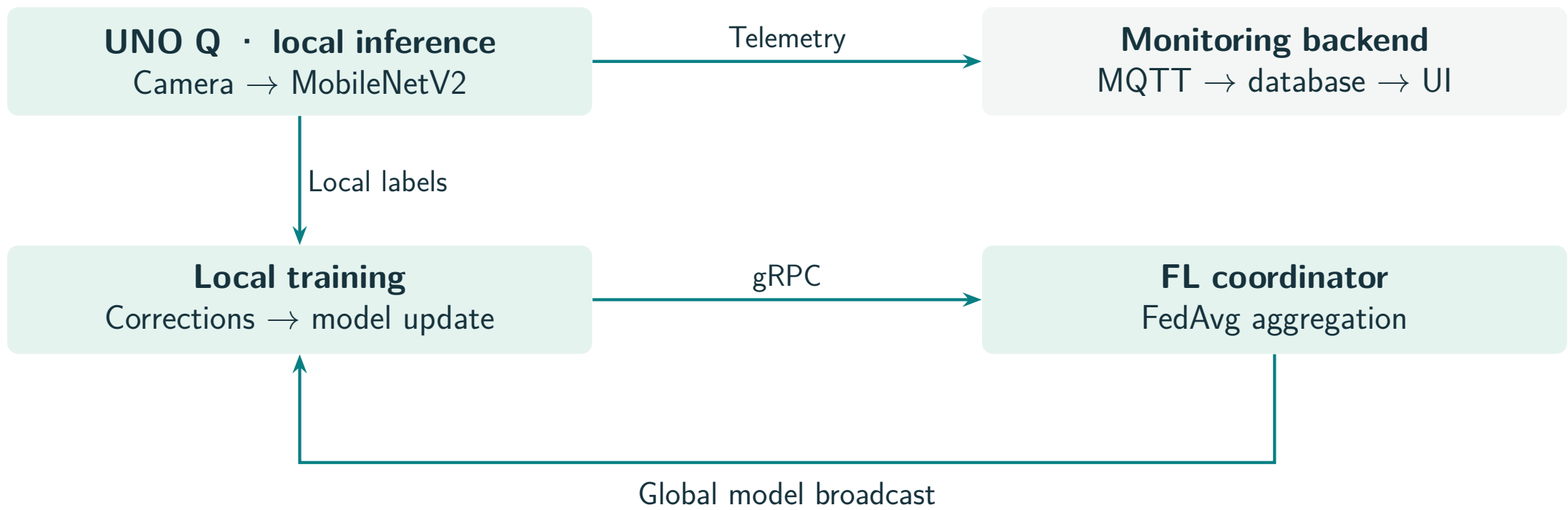
01 DISTRIBUTED INTELLIGENCE, LOCAL DATA

How federated learning works

Federated learning trains a shared model across devices while keeping images local. TrashUQ uses Arduino UNO Q boards to classify four waste categories with MobileNetV2.

Each device downloads the global model and fine-tunes its classification head using local user corrections, keeping the backbone frozen. It then sends a model update via **gRPC**.

The coordinator combines updates with **FedAvg** and returns the global model for the next cycle. **MQTT** carries telemetry separately.



2 edge nodes

UNO Q deployment
Hardware validation

93.75%

Accuracy with 20 clients
Synthetic-data simulation

44%

Less communication
4-bit stochastic rounding

02 HARDWARE VALIDATION

End-to-end operation

MQTT records	282
Successful aggregations	25/30
Median submission latency	~25 ms
Global model version	0 → 26

Concurrent-update handling

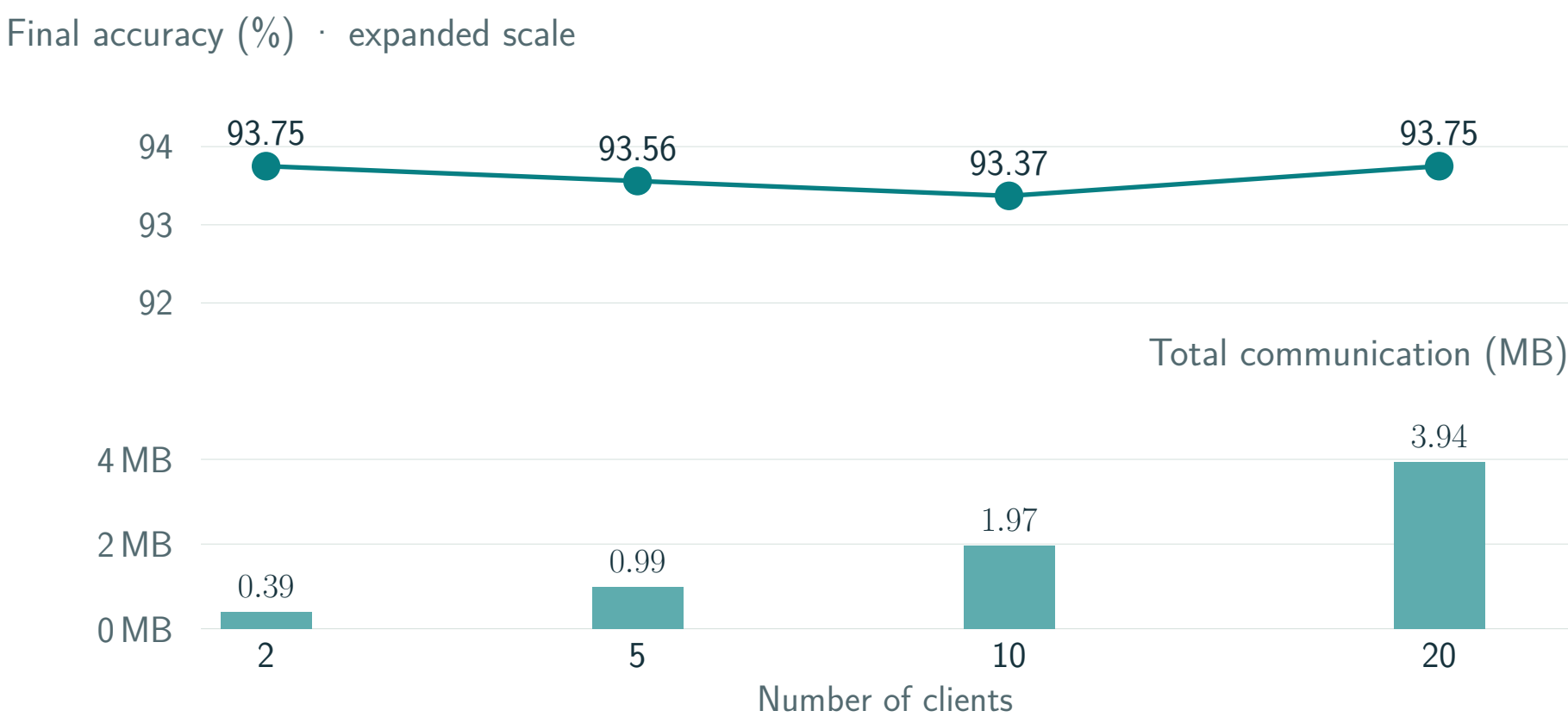


25 accepted / 5 stale updates rejected

12-minute session; simulated camera input and injected labels.

03 SCALING TO 20 CLIENTS

Stable accuracy, rising traffic



25 rounds · 3 seeds · synthetic data
Non-IID partitions: Dirichlet $\alpha = 0.3$.

04 COMMUNICATION-ACCURACY TRADE-OFF

4-bit updates preserve accuracy with 44% fewer bytes

20 clients · 25 rounds · means across 3 seeds · six compression methods

Method	Total communication (MB)	Accuracy	Byte savings
FP32	3.922	93.75%	0%
SR 8-bit	2.455	93.75%	37.4%
SR 4-bit	2.210	93.75%	43.7%
Top-50%	3.925	93.56%	-0.1%
Top-10%	2.357	84.66%	39.9%
Top-1%	2.003	63.83%	48.9%

SR = stochastic rounding. Top-k also transmits indices; aggressive sparsification reduces accuracy.

Conclusion: TrashUQ demonstrates federated learning on two validated edge nodes. In simulation, 4-bit stochastic rounding preserves accuracy while reducing communication by 44%.

Evidence: UNO Q deployment and synthetic-data FL experiments.
Simulation results do not establish real-world waste classification accuracy.

